Colorado State University

MECH 527: Hybrid Electric Vehicle Powertrains Sustainable Transportation

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Instructor Introduction

Thomas Bradley, Woodward Professor, Department Head of Systems Engineering

- Dr. Tom Bradley conducts research and teaches a variety of courses in system engineering, multidisciplinary optimization, and design. Dr. Bradley's research interests are focused on applications in Automotive and Aerospace System Design, Energy System Management, and Lifecycle Assessment. Dr. Bradley has coauthored 80+ peer reviewed archival papers, including seminal contributions to Model-Based Systems Engineering, fuel cell aviation, the design and development of plug in hybrid electric vehicles, and the lifecycle assessment of biofuels. He is an active member of the academic community as author of ~200 conference publications and academic reports. Dr. Bradley has served as PI and Co-PI for $24M in research funding since 2008.

Resources

- Google Scholar , Research Website

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Overview

- Sustainable Transportation
  - Definitions and Metrics
  - Mass versus Personal Transportation
  - Vehicle Lifecycle Thinking
  - Regulations that Control Transportation
- Summary and Discussion

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Sustainable Transportation Definitions and Metrics

- Sustainable transportation is defined as follows:
- In the natural environment:
  - **limit emissions and waste that pollute air, soil and water within the urban area's ability to absorb/recycle/cleanse;**
  - **provide power to vehicles from renewable or inexhaustible energy sources such as solar power in the long run**
  - **recycle natural resources used in vehicles and infrastructure**
- In society:
  - **provide equity of access for people and their goods, in this generation and in all future generations;**
  - **enhance human health; help support the highest quality of life compatible with available wealth; facilitate urban development at the human scale; limit noise intrusion below levels accepted by communities; and be safe for people and their property.**
- In the economy:
  - **be financially affordable in each generation; be designed and operated to maximize economic efficiency and minimize economic costs; and help support a strong, vibrant and diverse economy.**

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Sustainable Transportation Definitions and Metrics

- Societal
  - Portion of transport costs subsidized by public
  - Land area given over to transportation
  - Personal transportation expenses
- Environmental
  - Greenhouse gas emissions
  - Per capita transportation energy use
  - Criteria pollutants
  - Noise levels
  - Rate of Non-renewable energy consumption / rate of alternative consumption
- Cultural/Equity Related
  - Proportion of population with public transit within 500m
  - % of residential population exposed to road noise

Jeon and Amekudzi, 2005

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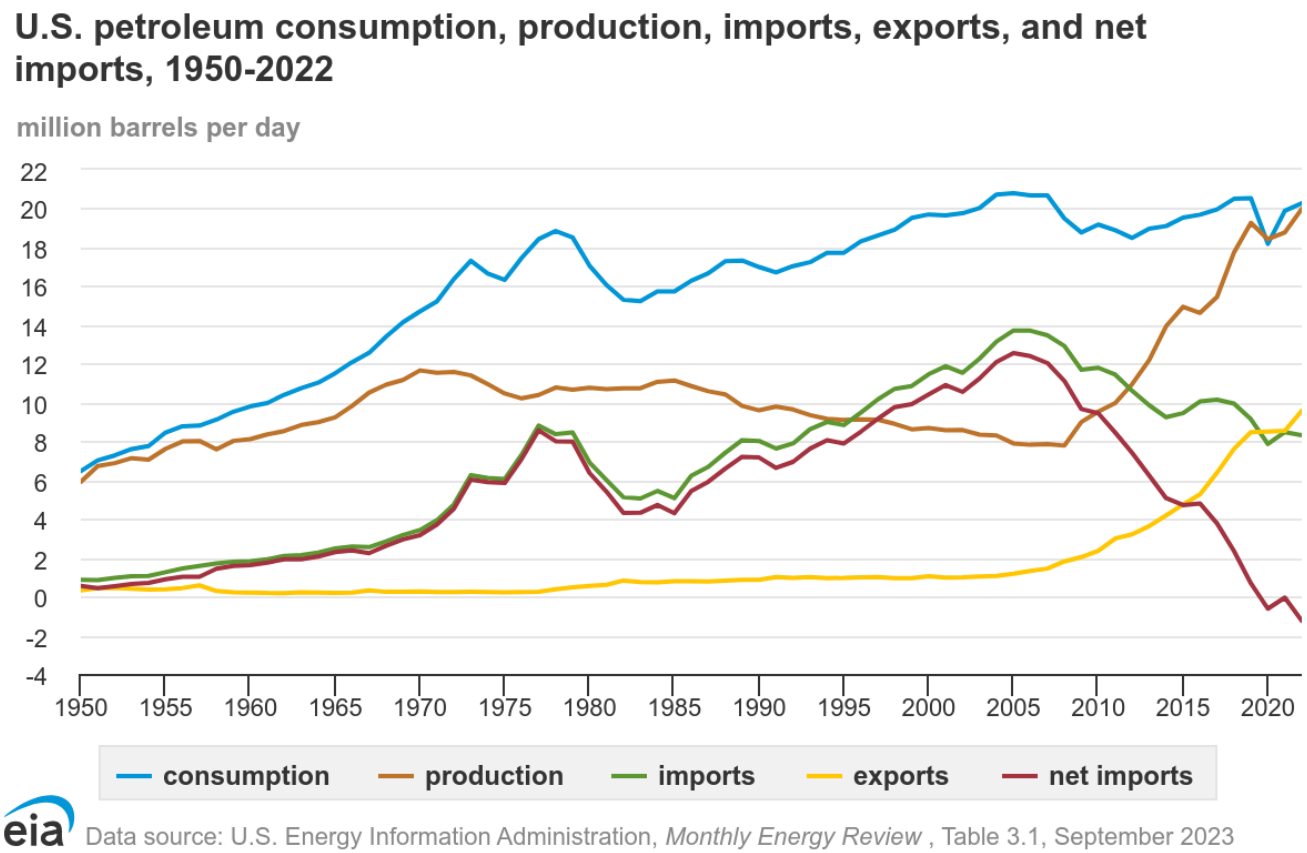
The (Un)Sustainability of Transportation

1) Petroleum Consumption



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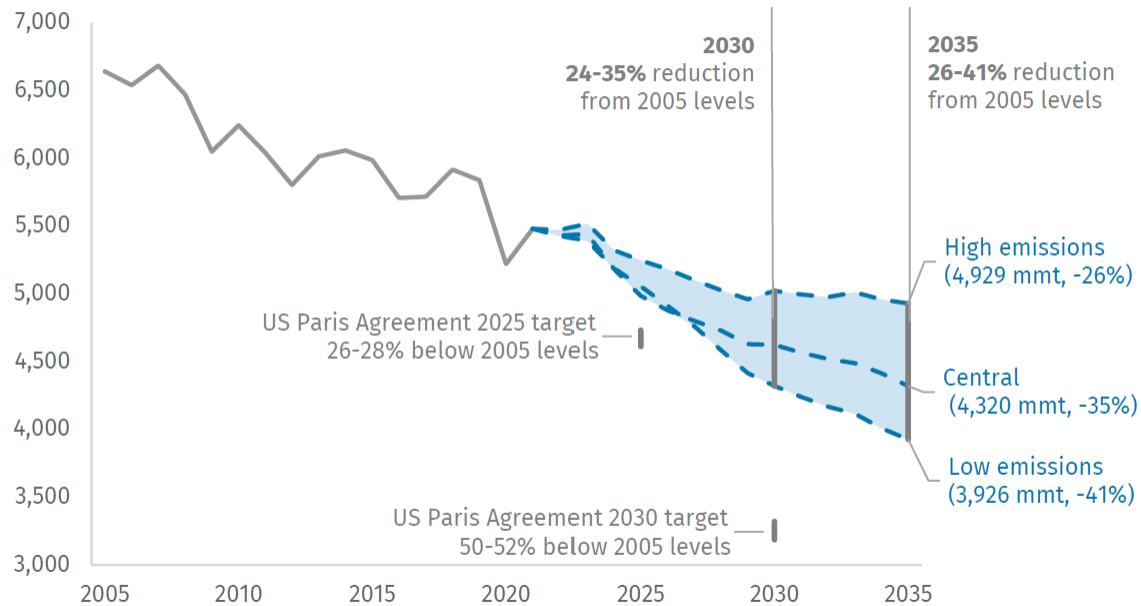
The (Un)Sustainability of Transportation

2) Greenhouse Gas Emissions

FIGURE 1

US greenhouse gas emissions under current policy

Net million metric tons (mmt) of CO_2e



Source: Rhodium Group

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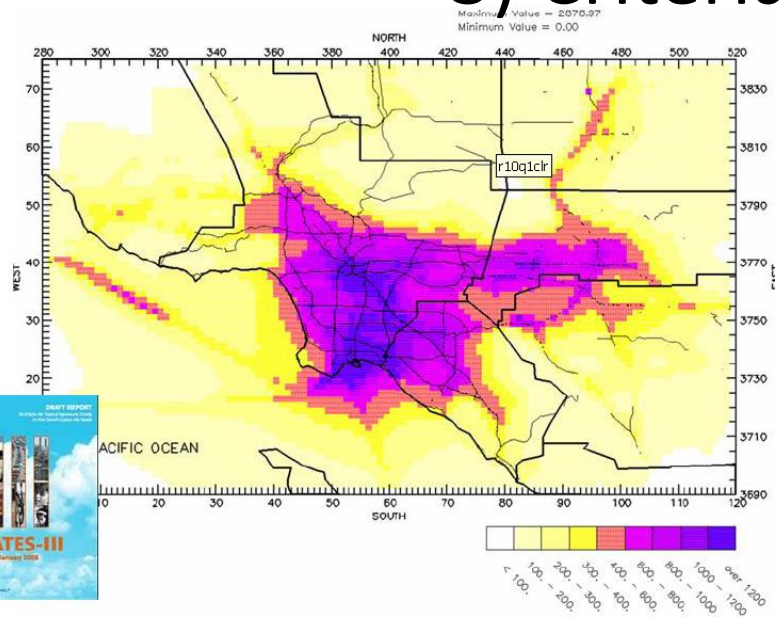
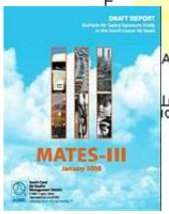
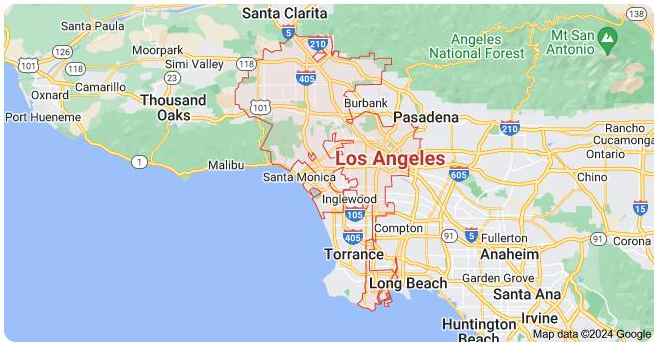
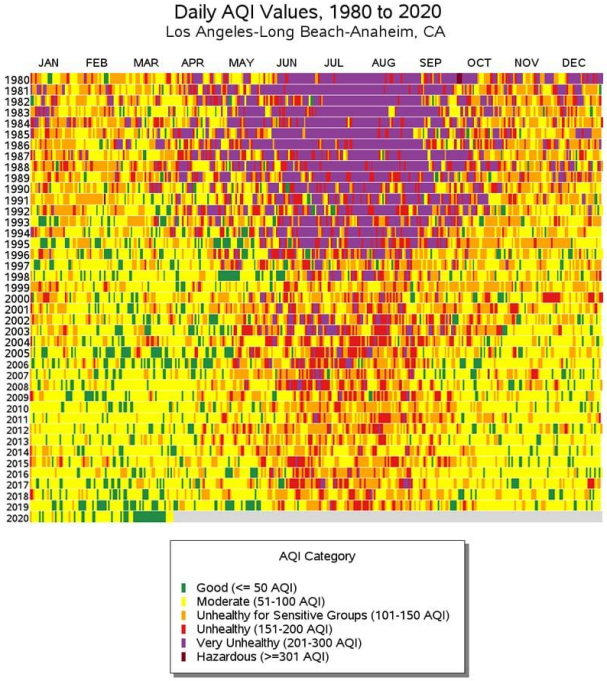
The (Un)Sustainability of Transportation

3) Criteria Air Pollutants









Los Angeles

90% of air pollution cancer risk is transportation related
Denver is in non-attainment

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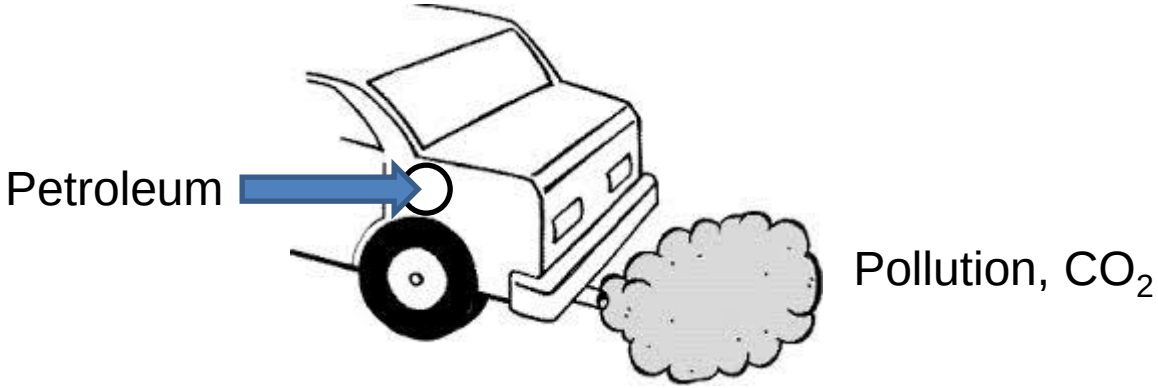
The (Un)Sustainability of Transportation



- As a society, we would like to improve the sustainability of transportation, but …
  - Transportation demand is increasing
  - The scope of the problem is huge
  - Automotive markets are hard to crack
  - New vehicles demand new infrastructure

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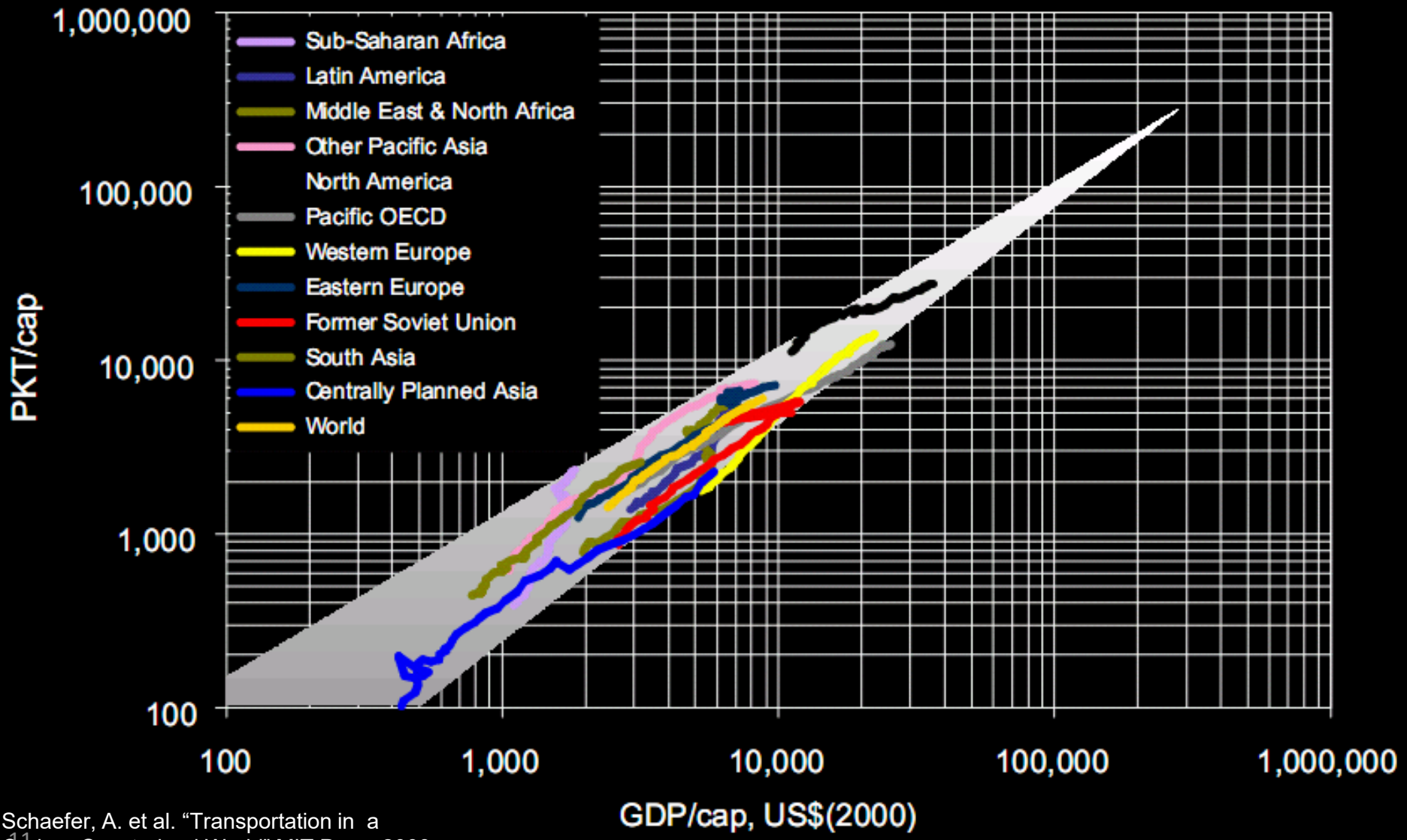
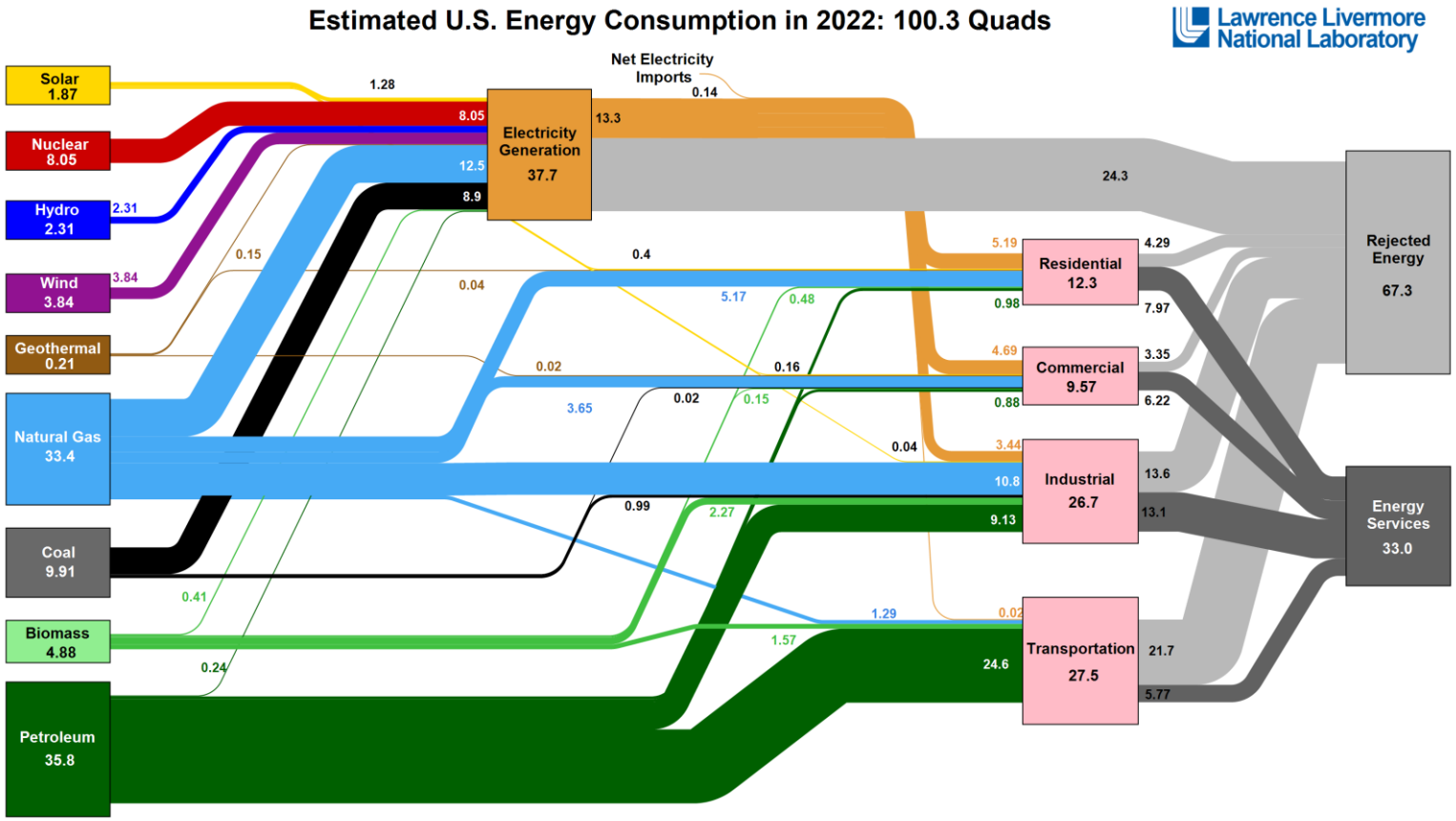
Transportation Demand is Increasing



Schaefer, A. et al. "Transportation in a Carbon Constrained World" MIT Press 2009.

The scope of the problem is huge



Source: LLNL July, 2023. Data is based on DOE/EIA SEDS (2021). If this information or a reproduction of it is used, credit must be given to the Lawrence Livermore National Laboratory and the Department of Energy, under whose auspices the work was performed. Distributed electricity represents only retail electricity sales and does not include self-generation. EIA reports consumption of renewable resources (i.e., hydro, wind, geothermal and solar) for electricity in BTU-equivalent values by assuming a typical fossil fuel plant heat rate. The efficiency of electricity production is calculated as the total retail electricity delivered divided by the primary energy input into electricity generation. End use efficiency is estimated as 0.65% for the residential sector, 0.65% for the commercial sector, 0.49% for the industrial sector, and 0.21% for the transportation sector. Totals may not equal sum of components due to independent Rounding. LLNL-MI-410527

Science 9 February 2007: vol. 315 no. 5813 796-798

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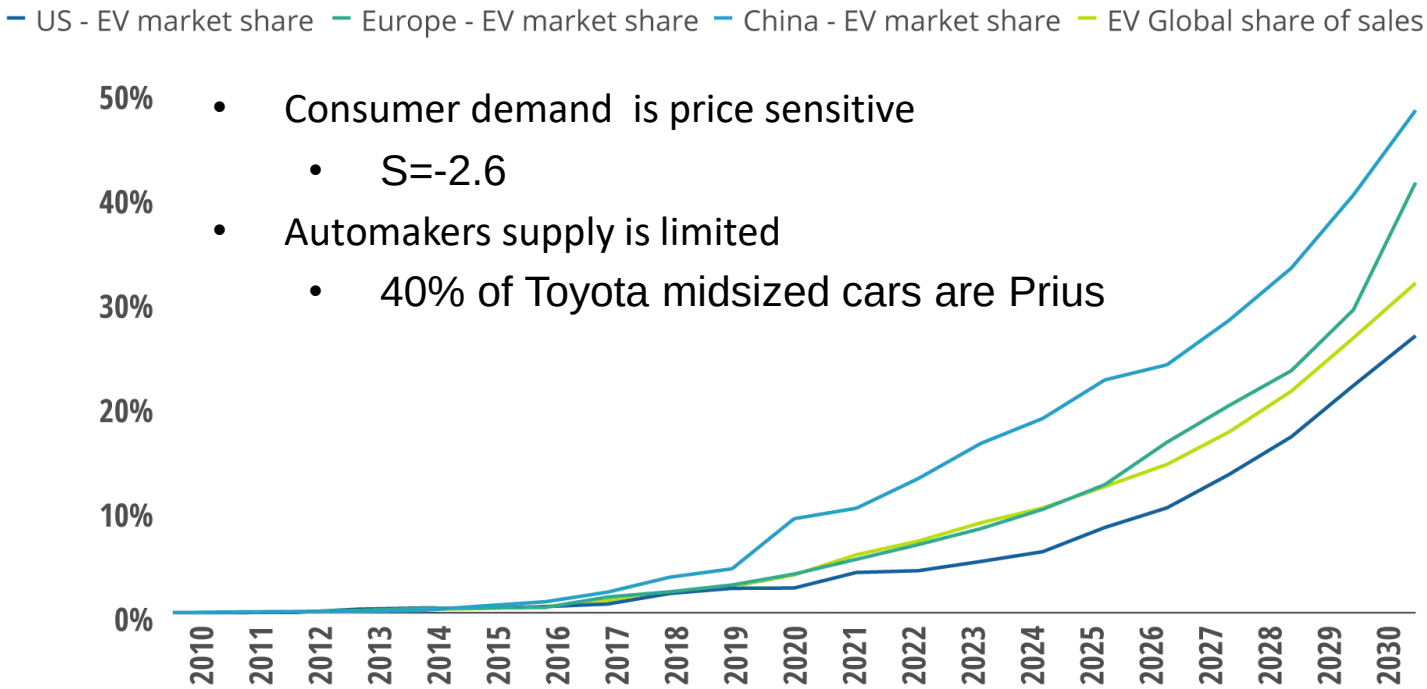
Auto. markets are hard to crack

FIGURE 3

Outlook for EV market share by major region



- Consumer demand is price sensitive
  - S=-2.6
- Automakers supply is limited
  - 40% of Toyota midsized cars are Prius

Source: Deloitte analysis, IHS Markit, EV-Volumes.com[17]

Deloitte Insights | deloitte.com/insights

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New vehicles demand new infrastructure

- New infrastructure can include
  - Fueling infrastructure
  - Product supply lines
  - Training of technicians and engineers
  - Safety regulations

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How to Improve Sustainability (GHG example)

$$GGE = \frac{GGE}{E}\frac{E}{VMT}\frac{VMT}{PMT}PMT$$

Fuel Substitution

Vehicle Engineering

Sociologists,
Policy Makers,
City Planners,
&
YOU! and ME!

GGE=A unit of greenhouse gas emissions
E=A unit of energy
VMT=A unit of vehicle distance travelled
PMT=A unit of personal travel distance

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Fuel Substitution

- Biofuels (Ethanol, biodiesel)
- Electricity
- Hydrogen
- Natural Gas

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Fuel Substitution – Electricity

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Fuel Substitution – Electricity

- Is fueling transportation with electricity good or bad?
- We must use Lifecycle Assessment to find out…

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Vehicle Engineering

- Characteristics of HEV Drivetrains (In approximate order of importance to efficiency)
  - Isolation of Engine from Vehicle Operating Conditions
  - Regenerative Braking
  - Engine Downsizing
  - Electric Drive Mode
  - Electrified Accessories
  - No Engine Idle
  - Energy Banking

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Mass versus Personal Transportation

- Energy Consumption of Mass Transit
  - Heavy Commuter Rail = 171-360 ACWh/person/mi
  - Nissan Leaf EV = 300 ACWh/mi

  - 747-400 cruise = 70 miles/passenger/gallon kerosene
  - Greyhound bus = 162 miles/passenger/gallon diesel
    - LHV of Diesel is ~11kWh/L
    - LHV of Gasoline is ~10kWh/L

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Transportation Futures

- How do we get more sustainable vehicles on the road?
  - Wait for gasoline prices to rise and technology prices to decline
  - Engineer BETTER vehicles
  - Policy
    - Corporate Average Fuel Economy
    - California ZEV Program
  - Research and Education

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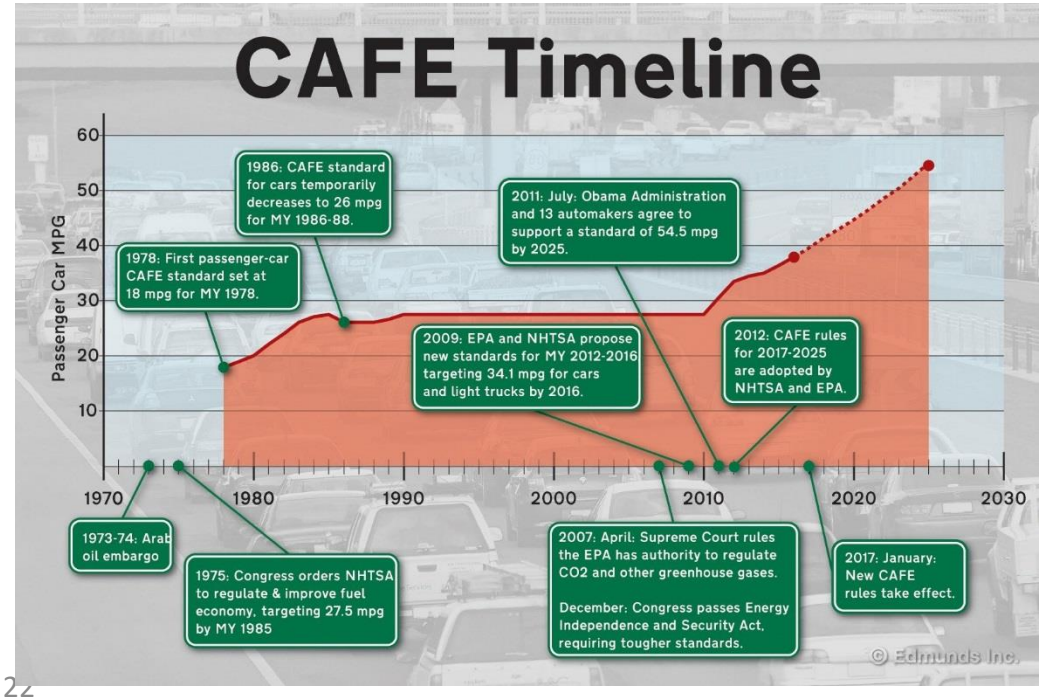
Policy Drivers for Sustainable Transportation

- **Corporate Average Fuel Economy (EPAct)**
  - The average fuel economy of the entire corporation's fleet
  - Every MPG increase costs each automaker ~$1B
  - To date automakers have met these requirements with
    - Electrified accessories, cylinder deactivation, high efficiency tires, HEVs, etc.
    - EVs and PHEVs will play a role in the future



- **State Incentives**
- **Colorado**
- CO offers an income tax credit of $5,000 for those that purchase an EV and $2,500 for those who lease an EV.
- BEVs and PHEVs are eligible for the HOV lane with an application.
- PEVs are exempt from emissions testing, but must pay a $50 annual registration fee according to HB 1110, Section 12, 42-3-304.
- Charge Ahead Colorado offers funding for up to 80% of the cost for EVSE, up to $3,260 for a level 2 station with a single port and $6,260 for a multi-port, and $13,000 for a level 3 station with a single port and $16,000 for a multi-port.

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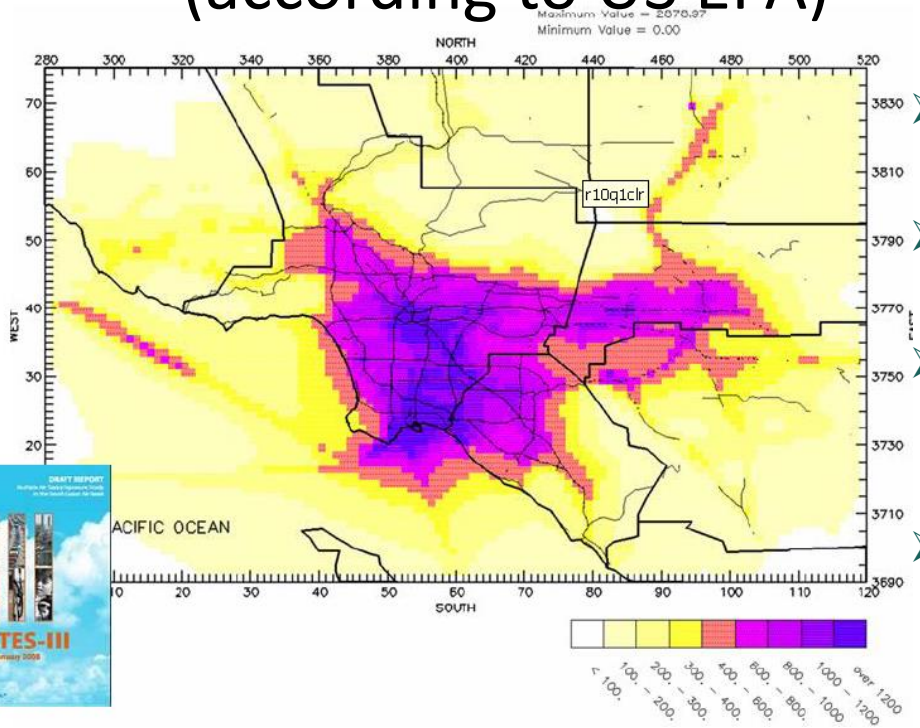
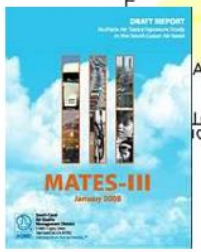
California ZEV Program

- California is the only state which is allowed to set its own vehicle emissions standards
- Why? 80% of Californians live in unhealthy air (according to US EPA)





- **Required 2% ZEV sales in California by 1998 (1990)**
- **Modified to require a good faith effort (1996) and high technology vehicles**
- **Modified to allow ZEV demonstrations and SULEV gasoline vehicles (2003)**
- **Zero-emission or plug-in hybrid vehicles will account for one in seven new cars sold in California in 2025 (15.4 percent).**
- **More than 1.5 million zero-emission and plug-in hybrid vehicles will be on the road in California by 2025.**

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Summary and Discussion

- We are going to learn more about many of these aspects over the course
  - Vehicle Simulation
  - Vehicle Engineering
  - Alternative Fuels
  - Lifecycle Assessment
  - Systems Thinking on Sustainability/Transportation/Energy
- Discussion Points

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